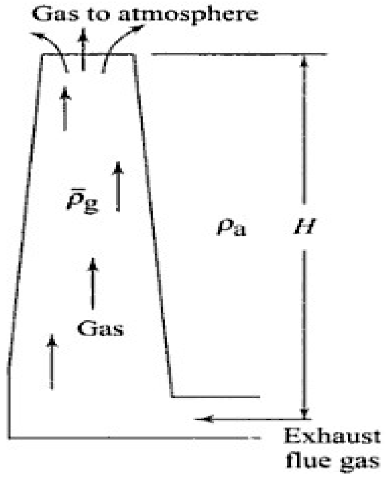
Q2

The natural draught is produced by a chimney or a stack. It is caused by the density difference between the atmospheric air and the hot gas in the stack. For a chimney of height H metres (Fig. 4.14) the draught or pressure difference (N/m^2) produced is given by

$$\Delta p = gH\,(\rho_a - \bar{\rho}_g) \qquad \mathbf{1}$$



Draught produced by a chimney

where ρ_a = density of atmospheric air, kg/m^3
ρ_g = average gas density in the chimney, kg/m^3
g = acceleration due to gravity, 9.81 m/s^2

Let us assume that the volume of the products of combustion is equal to the volume of air supplied for combustion, both volumes being measured at the same temperature. Thus, the volume of 1 kg of flue gas at NTP

$$v_a = \frac{0.287 \times 273}{101.325} = 0.7733\ m^3/kg$$

The volume of m kg of air per kg fuel at temperature T_a

$$V_a = 0.7733\ m \times \frac{T_a}{273}$$

$$\rho_a = \frac{1}{0.7733} \times \frac{273}{T_a} = 1.293\left(\frac{273}{T_a}\right)\ kg/m^3$$

The mass of flue gas will be $(m + 1)$ kg and its temperature is T_g.

Density of flue gas, $\rho_g = \dfrac{m+1}{0.7733 \times m \times T_g/273}$

$$= 1.293\left(\frac{273}{T_g}\right)\left(\frac{m+1}{m}\right) = \frac{353}{T_g}\left(\frac{m+1}{m}\right)$$

∴ Draught produced (from Eq. 1)

$$\Delta p = 1.293 \times 273\left[\frac{1}{T_a} - \frac{m+1}{m}\frac{1}{T_g}\right]gH$$

$$= 353\, gH\left[\frac{1}{T_a} - \frac{m+1}{m}\frac{1}{T_g}\right] \qquad 2$$

If the draught is measured in terms of water column and the value is h mm,

$$\Delta p = 10^3\, gh \times 10^{-3} = 353\, gH\left[\frac{1}{T_a} - \frac{m+1}{m}\frac{1}{T_g}\right] \qquad 3:$$

∴
$$h = 353\, H\left[\frac{1}{T_a} - \frac{m+1}{m}\frac{1}{T_g}\right]$$

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where h is in mm and H is in m.

For given H and T_a, the natural draught depends on the average gas temperature T_g. Higher is the T_g, higher is the draught produced. But a high T_g means a large exhaust loss through chimney resulting in a lower boiler efficiency. With an optimum T_g, the amount of draught produced by density difference is thus limited.